

Total No. of Printed Pages: 02

**SUBJECT CODE NO: - RNP-8010**

**FACULTY OF SCIENCE AND TECHNOLOGY**

**S.E. (Civil Engg.) (NEP) Sem - III**

**Examination November / December 2025 (To be held in February-2026)**

**Surveying**

**[Time: 2:00 Hours]**

**[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B

1. **Part A: Question No 1 is compulsory.**
2. **Attempt any four questions from remaining Part B: Q.2 to Q.7.**

**Part A**

**Q1 Choose the correct alternative for the following questions:**

**10**

- 1) Levelling is used to determine:
 

|                         |                       |
|-------------------------|-----------------------|
| (a) Horizontal distance | (b) Vertical distance |
| (c) Angle between lines | (d) None              |
- 2) An instrument used for levelling is:
 

|                |                 |             |                |
|----------------|-----------------|-------------|----------------|
| (a) Theodolite | (b) Dumpy Level | (c) Compass | (d) Clinometer |
|----------------|-----------------|-------------|----------------|
- 3) A temporary benchmark is:
 

|                          |                             |
|--------------------------|-----------------------------|
| (a) Fixed permanently    | (b) Established temporarily |
| (c) Used for all surveys | (d) None                    |
- 4) Contour lines of different elevations never:
 

|                               |                  |
|-------------------------------|------------------|
| (a) Cross each other          | (b) Merge        |
| (c) Intersect at water bodies | (d) Are parallel |
- 5) Reciprocal levelling is used when:
 

|                                     |                           |
|-------------------------------------|---------------------------|
| (a) Instrument is in good condition | (b) Ground is uneven      |
| (c) Long sights are taken           | (d) For slope measurement |
- 6) Auto level is a type of:
 

|                   |                               |
|-------------------|-------------------------------|
| (a) Digital level | (b) Self-levelling instrument |
| (c) Theodolite    | (d) Compass                   |
- 7) Levelling staff is graduated in:
 

|          |            |            |           |
|----------|------------|------------|-----------|
| (a) Feet | (b) Inches | (c) Metres | (d) Yards |
|----------|------------|------------|-----------|
- 8) Back sight is the reading taken on:
 

|               |                   |
|---------------|-------------------|
| (a) Benchmark | (b) Turning point |
|---------------|-------------------|

- (c) Fore station                      (d) Intermediate point
- 9) Contour lines close together indicate:  
 (a) Flat slope                      (b) Steep slope  
 (c) Uniform slope                      (d) Depression
- 10) The process of checking staff readings is known as:  
 (a) Booking                      (b) Fly levelling  
 (c) Check levelling                      (d) Reciprocal levelling

### Part B

**Solve any FOUR of the following**

**20**

- Q2** Explain with neat sketch the types of levelling. **05**
- Q3** Describe linear and angular methods of setting out simple circular curves. **05**
- Q4** Write the advantages and disadvantages of Plane table Survey. **05**
- Q5** Explain the characteristics and uses of contour lines. **05**
- Q6** The following staff readings were observed successively with a level the instrument is moved by third sixth and eighth readings. 2.228 :1.606 :0.988 :2.090 :2.864 :1.262 :0.602 :1.982 :1.044 :2.684 m enter the reading in record book and calculate R.L. if the first reading was taken at a B.M of 432.383m **05**
- Q7** Describe the testing and permanent adjustments of Dumpy Level. **05**